

## Experiment Number (7)

### Starch & Casein Hydrolysis

Talk about the use of microorganisms in life which is their ability to

Produce certain enzymes. From  $\begin{cases} \text{Plant} \\ \text{Animal} \\ \text{Microbial} \end{cases} \rightarrow \text{Prepared in industrial.}$   
biological catalyst that speed up the reaction by lowering its activation energy without being consumed.

enzyme vs Chemical Catalyst in industrial?

because of: ① Biological system help in ingredient substitution.

② less undesirable Product

③ Increased Product yields

④ Less energy intensive

⑤ Less Polluting

Why Microbial enzyme are Prepared?

easy, Cost effective, More stable, High Production, ease to genetic manipulation.

#### $\alpha$ -amylase

degrade  $\alpha$ -1,4 glycosidic bond (starch)

- Starch Processing.
- Detergent industry.
- Bakery & anti-salting industry.
- Food industry.
- Paper industry.
- Textile industry.
- Alcohol industry.
- Pharmaceutical industry.

#### Caseinase

- exoenzyme
- degrade casein  $\rightarrow$  Peptone
  - $\downarrow$
  - $\downarrow$
  - $\downarrow$
  - Dipeptide  $\alpha$ -a Polypeptide
- used in dairy Product

LAB Work: check the ability for bacteria to Produce the enzyme of interest (Biochemical test)

\*\* Put the substrate with the media (starch) & check the bacteria that grow for the presence of enzyme

( $\alpha$ -amylase)

we use Iodine to test the Presence of ~~starch~~

N.A + starch Iodine  $\rightarrow$  blue color

N.A + Iodine  $\rightarrow$  brown color (light brown)

السكر  $\alpha$ -amylase في السكر  
No blue color  $\leftarrow$  No starch  
+ve result (zone of hydrolysis)



## Caseinase activity

\*\* substrate added to media (milk as a source of substrate)  
↑  
rich in casein

Skim milk + NA (Pour Plate method)

↓  
inoculation in wells → incubate

↓  
+ve result → clear zone around bacterial growth  
↓  
caseinase activity

we can differentiate bacteria according to the diameter of the clear zone.

(Qualitative & Quantitative)

|            | $\alpha$ -amylase                  | caseinase       |
|------------|------------------------------------|-----------------|
| Substrate  | starch                             | casein          |
| Medium     | starch agar                        | skim milk agar  |
| incubation | 24 hour                            | 24 hour         |
| indicator  | Iodine                             | Opacity         |
| +ve result | clear zone around the growth       | loss of opacity |
| -ve result | blue-black color around the growth | opacity.        |



## Experiment number (8)

### Milk quality, Yoghurt & Cheese

• test for contamination, source of contamination, way of Processing, etc

Source of contamination:

- ① Udder of cow      ② handling

Type of bacteria that cause contamination: (PAA)

G-ve, pathogenic, cyclo-trophs  
نظارة منفردة

Pseudomonas, Acinetobacter, Alcaligenes.

Pasteurization: almost

Kill of all of the spoilage organisms.

Pasteurization method ~~choose~~ depends on:

- intended purpose of pasteurization
- volume of milk to be pasteurize
- target microorganism

↓  
active ~~نظارة~~

(~~نظارة~~) they produce  
Proteases & lipases.

| Method                                   | Temperature  | Time                                   | Target                                                                      |
|------------------------------------------|--------------|----------------------------------------|-----------------------------------------------------------------------------|
| Normal                                   | 68°C         | 30 min                                 | Pathogenic bacteria                                                         |
| Flash Pasteurization<br>(HTST)           | 72°C to 74°C | 15-20 <sup>second</sup> <del>min</del> | resistance pathogenic bacterial<br>spores (Clostridium botulinum<br>spores) |
| Ultra high<br>temperature past.<br>(UHT) | 135°-140°C   | 2-4 second                             | Coxiella burnetii (cause Q fever)                                           |

How to study them?

- standard plate count
- Mosely Method
- reductase test (reduction)

electron transport chain

مع استخدام مادة نقد، نشوف التغيير على  $e^-$  تستقبل electrons وينتهي به

Final electron acceptor ~~نظارة~~



## Yoghurt Formation

Lactose

- streptococcus thermophiles
- lactobacillus acidophilus
- lactobacillus bulgaricus

→ lactic Acid

↳ denaturation of casein to form  
Fibrous of casein protein  
(responsible for yoghurt texture)

## LAB Work

### Quality of Milk

Milk + methylene blue

→ electron acceptor

→ leukomethylene

Oxidized

Reduced

blue → Oxidized

colorless → Reduced

milk + methylene blue → blue

✓ GOOD

milk + methylene blue → white

✗ BAD SAMPLE

### Yoghurt Formation

milk + teaspoon of

→ source of bacteria  
→ yoghurt

milk + lactobacillus → yoghurt

## Cheese Making

Rennet: complex of enzymes like chymosin, Pepsin, lipase.

↓  
Protease degrade casein

Form network

traps other  
component of milk like Fat, minerals

Cheese

CaCl<sub>2</sub>: improve rennet  
coagulation  
Process.



# Honey & clorex as antimicrobial Agents.

Resistance Bacteria species!!

- Natural Anti-microbial Agents

- Microbs that Produce Anti-microbial agent.

- Bacterial transformation

→ WHY NATURAL?!

• side effect of it is less or NO side effect!

نظراً، من الفاعلية متوازنة (سواءاً الشفاء)

• have different mechanisms of action.

مثل: العسل أو الثوم

Honey as an anti-microbial agent

\*it is also antioxidant agent

→ bacterial  
→ fungal

حماية كبيرة جداً من السرطان في تحمية تليق به السائل المنوي

• ما هو العسل؟

38% ~~Fructose~~

Fructose, maltose, glucose.

- saturated or super-saturated solution of sugars 84% from dry weight

- contain vitamins, enzymes, Proteins

What makes Honey an Anti-microbial Agent?!

• contain low amount of water & high amount of sugar which result in a very small amount of water available for the bacteria

(aques activity) البكتيريا تحتاج كمية الجزيء الماء

Free water activity → Humidity رطوبة

Preservation كما قل المحتوى المائي؛ اذ أن قسراً حفظاً لا يفسد

0.6-0.62 → water activity for the honey

0.94-0.99 → bacteria require for growth

water activity



الانقلاط بافتلاف العطاء النباتي الى ارض منه الرعيه لتقوية العسل

Acidity

3.2 - 4.5

NOT Suitable For bacterial growth (need neutral PH)

\*  $H_2O_2$  Catalase

\* كيف يصنع العسل؟

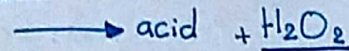
خلية العسل تتكون من اقراص شمعية (شمع + صلبون لؤلؤ) تصنعها النمل على شكل عيون سداسية (مخار zero) مبطنة دمم للعيون السداسية الناشئة، ما تبين الملكة في قاع العيون السداسية بعدما تقول النمل بعد الى يرسة ثم شرنقة ثم خلية.

يستخدم النمل هذه العيون للعسل.

الرعيه قلوب مخفف 10-15% سكر، تاخذ النمل بواسطة الخرطوم الخاص الى معدة خاصة غير معدة الجمال

المضغ، وحينها هذه المعدة معالجة الرعيه enzymatically.

تتوزع النمل في  
glucose oxidase  
عشاء يتغل



sterilizing agent

hypopharyngeal glands

ثم يتغل عبر الخرطوم الخاص من معدة النمل الى اخرى واخرى... وتتم معالجة كل مرة

حتى يصل الى اكمال المعالجة، تصنع النمل في العيون السداسية.

النمل يتلش بقل تيار، هواء عشاءه تقلل المحتوى المائي للعسل وبالتالي يستمر محفوظ لمدة أطول.

MIC: minimum inhibitory concentration

كما كان ال conc. أقل فهو أكثر effective

LAB Work:

تحفيان مختلفه من العسل

0%, 1%, 5% Honey

نفرين بكثيرة

و conc. مختلفه من الكلور

ب Plate بال wells

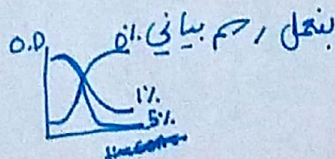
تاخذ القراءات لل optical density باستخدام

spectrophotometer

• مرات (بالخافضة)

0 hr, 1 hr, 2 hr, 24 hr, 48 hr

O.D



بدى اقراصا  
مطبوعا على اعدادات

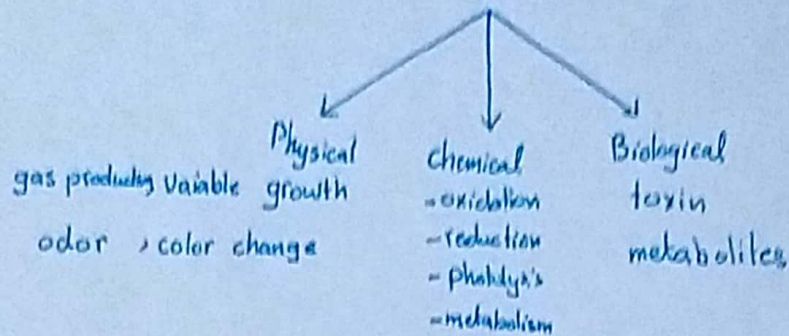


# Microbial Spoilage of Pharmaceutical Products.

Spoilage: Deterioration of the Product (chemical/Physical changes)

Types:

- ① Physical: heat, evaporation ...
- ② Chemical: oxidation, reduction, Photolysis
- ③ Microbial: contamination with bacteria or Fungi (yeast/mold)



Source of contamination:

- ① Inadequate preparation
- ② Improper sterilization
- ③ Improper storage
- ④ Accidentally exposure to the environment.

\*\* To prevent microbial contamination → Preservatives

## LAB Work

• enumeration

- Plate count Agar ✓

- most Probable number

- optical density

• identification

